Network Modeling and Analysis in Python

Today's goal • 3/3 items

Module 1
Connectivity in Networks

Welcome to the Course
- Welcome to Network Modeling and Analysis in Python — Video • 5 min
- MADSwPy Certificate Roadmap — Reading • 10 min
- **Course Syllabus** — Reading • 10 min
- Introduction to Jupyter Notebook — Reading • 10 min
- Meet Other Learners — Discussion Prompt • 10 min
- Help Us Learn About You — Reading • 10 min

Homophily and Assortativity
- Homophily and Assortativity — Video • 16 min

Structural Holes
- Structural Holes — Video • 12 min

K-core Decomposition
- K-core Decomposition — Video • 8 min
- Knowledge Check: Connectivity in Networks — Practice Assignment • 35 min

Module 1 Wrap-Up
- Demo: Connectivity NetworkX Tutorial — Video • 6 min
- Module 1 NetworkX Tutorial — Lab • 1h
- Module 1 Assignment — Graded Assignment • 30 min
- Module 1 Programming Assignment: Connectivity in Networks — Programming Assignment • 3h
- Optional Readings and Resources — Reading • 10 min

Optional Lesson: Review of Key Concepts
- Review of Key Concept — Reading • 10 min
- Network Definitions and Vocabulary — Reading • 10 min
- Network Connectivity — Reading • 10 min
- Influence Measure and Network Centralization — Reading • 10 min
- Network Evolution — Reading • 10 min
- Optional-Review of Key Concepts — Practice Assignment • 30 min

Module 2
Community Structure in Networks

Module 3
Network Generative Models

Module 4
Models of Diffusion in Networks

Course Syllabus

0:39 / 12:31 2x

Welcome!

Welcome to Network Modeling and Analysis in Python, taught by Daniel Romero!

In a world where everything is connected, understanding how networks function is a key skill for any data scientist. In this course, you'll learn how to analyze, model, and interpret network data using Python. You'll explore real-world datasets and apply techniques to uncover patterns such as homophily, structural holes, and community structures. You will also explore generative models for simulating networks and study diffusion processes, where you will learn how to identify the most influential nodes within a network.

This is the third course in the **More Applied Data Science with Python** specialization from the University of Michigan. We recommend that you take this course after completing Data Mining in Python and Applied Unsupervised Learning in Python. We also suggest that before taking this course that you complete the **Applied Social Network Analysis in Python** course from the Applied Data Science with Python (ADSwPY) series.

Participation Strategies

Engaged learning looks different for everybody. In this course, we hope you will define your own measures of success and engage with the material in a way that best suits your needs.We recognize and celebrate the diverse ways learners engage in courses. As you go through this course, we hope you will reflect on your unique skills, needs, and aspirations, and engage in the course material in a way that aligns with your own goals. While the course provides time estimates for completion, you should feel empowered to engage in the material in whatever ways make sense to you.

Course Structure & Grading Policy

The course is structured into four modules. In each module, we have included lecture videos, ungraded knowledge check assignments, auto-graded assignments, and programming assignments.

Module 1: Connectivity in Networks

In this module, you will learn about the foundational concepts and structural properties that characterize connectivity in networks when considering node attributes. You will explore the principle of homophily or assortative mixing, which explains the tendency of nodes to connect with others that are similar to themselves, and reciprocity, which addresses the mutual linkage between nodes. The module will also cover the concept of structural holes, which highlights the advantages of nodes positioned between unconnected network clusters, and the k-core decomposition method, used to identify cohesive subgroups within the network.

- Graded Assessment
  - Module 1 Graded Assignment: Connectivity in Networks (7%)
  - Module 1 Programming Assignment: Connectivity in Networks (18%)

Module 2: Community Structure in Networks

This module covers Community Structure in networks -- the organization of nodes in a network into clusters or communities, where nodes within the same community have a higher density of connections within their community than across other communities. We explore algorithms to identify communities in networks and evaluate them. Key topics include Modularity, a measure that quantifies the strength of the division of a network into modules or communities; the Girvan-Newman algorithm, a method that systematically removes edges from the network to find the best division based on edge betweenness centrality; Agglomerative Hierarchical Clustering, a technique that builds a hierarchy of clusters by progressively merging groups based on their distance or similarity; and Label Propagation, an algorithm for detecting communities based on spreading labels throughout the network and forming communities based on the dominant label. We also discuss applications to the community detection problem in real-world scenarios.

- Graded Assessment
  - Module 2 Graded Assignment: Community Structure in Networks (7%)
  - Module 2 Programming Assignment: Community Structure in Networks (18%)

Module 3: Network Generative Models

This module expands on network generative models, building on previously covered models such as Small-World and Preferential Attachment models. We'll explore the Erdős-Rényi model, which connects nodes randomly and serves as a baseline for understanding random graph theory. The module also covers the Stochastic Block Model, which is useful for modeling community structures by grouping nodes and connecting them based on group membership. Additionally, we explore the Configuration Model, which is used for creating random networks that maintain a given degree distribution.

- Graded Assessment
  - Module 3 Graded Assignment: Network Generative Models (7%)
  - Module 3 Programming Assignment: Network Generative Models (18%)

Module 4: Models of Diffusion in Networks

This module explores how ideas, diseases, and information spread in networks using models like SI, SIS, SIR, Independent Cascade, and Linear Threshold. Learners will simulate these models with Python, modify them, and tackle the influence maximization problem, identifying key nodes to optimize information or behavior spread.

- Graded Assessment
  - Module 4 Graded Assignment: Models of Diffusion on Networks (7%)
  - Module 4 Programming Assignment: Models of Diffusion on Networks (18%)

Community Engagement

We expect everyone to be mindful of what they say and its potential impact on others. The goal is to have respectful discussions that do not violate the community space created for these conversations. Here are some productive ways to engage in this course:

- **Participate:** This is a community. Read what others have written and share your thoughts.
- **Stay curious:** Learn from experts and each other by listening and asking questions, not making assumptions.
- **Keep your passion positive:** When replying to a discussion forum post, respond with thoughts on what was said, not about the person who posted. Avoid using all caps, too many exclamation points, or aggressive language.
- **Acknowledge discomfort:** The topics discussed in this course might be challenging or hard to talk about. Stick with it and remember the benefits of having these tough conversations that surface from multiple perspectives.

We expect all learners to abide by our full Learner Engagement Policy. We will specifically be monitoring this course for language that could be considered inflammatory, incivil, racist, or otherwise unacceptable for this learning space, and we will remove language deemed such.

Note regarding "study groups" outside of the platform: While learners are encouraged to interact with one another using communication tools offered within the learning platform, note that any study groups formed outside of this platform, such as over social media or communication applications and websites (e.g., WhatsApp) are not affiliated, endorsed, or moderated by the University of Michigan. If you receive an invitation to an outside study group mentioning the course and claiming to have any official connection to U-M or individual instructors, please exercise caution as this may be an attempted scam.

Academic Honesty

All submitted work should be your own and academic dishonesty is not allowed, such as, but not limited to:

- Copying words, ideas, or other materials from another source without giving credit to the original author
- Copying answers or copying from your peers within the course
- Employing or allowing another person to alter or revise your work, and then submitting the work as your own
- Using Artificial Intelligence tools, such as ChatGPT, to create or edit your work and submitting that work as your own, unless you were instructed to use AI as part of the assignment

Academic dishonesty may result in deleting assignment submissions, forum responses, removal from the course, or other actions as needed. For more information on academic honesty and misconduct, please refer to our Learner Engagement Policy.

Course Support

Questions and discussion of course material should take place within the course itself. Please do not contact instructors or teaching assistants off the platform, as responding to individual questions is virtually impossible.

We encourage you to direct your questions to Course Issues and Questions forum, where your question might be answered by a fellow learner or one of our course team members. For technical help please contact the Coursera Learner Help Center support forums.

Accessibility

We are committed to developing accessible learning experiences for the widest possible audience. We recognize that learners with disabilities (including but not limited to visual impairments, hearing impairments, cognitive disabilities, or motor disabilities) might need more specific accessibility-related support to achieve learning goals in this course.

Please use the accessibility feedback form to let us know about any accessibility challenges such as urgent issues that keep you from making progress in the course (e.g., missing or inadequate alt-text, captioning errors).

Third-party sites and software: While the University of Michigan is not responsible for the accessibility of third-party sites or applications that may be linked from this course, we still encourage you to report associated third-party accessibility issues so that we can ensure you are able to participate. In such cases, we may contact you for additional information as we investigate ways of removing accessibility barriers or to suggest accessible alternatives.

Global Learning Statement

We welcome all learners to this course. By enrolling in this course, you join people from all over the world and we value the perspectives they bring to the learning environment. We strive to create a community of mutual respect and trust where people from all backgrounds and views are valued and heard without the threat of bias, harassment, intimidation, or discrimination. We pay attention to your feedback on how different types of learners experience this course and aim to make improvements so the course can best serve everyone. We hope you enjoy learning about topics that are important to you.

Tools & Technology

Coursera Platform

If you are new to Coursera and you want to familiarize yourself with the platform, read the Help Articles in Coursera Help Center. This is a good resource where you can find answers to many basic questions such as how to adjust video settings, how to submit assignments, and how to earn a course certificate.

H5P

This course includes course content and learning activities built with a tool called H5P to include additional interactivity in this course. These activities will appear within the course itself, to avoid redirecting you outside of the learning platform. If you encounter challenges when exploring the H5P interactive elements, we recommend that you check the course discussion forums on Technical Support for assistance.

GenAI Tools

Please note that any independent use of GenAI tools to complete assignments remains subject to the Academic Honesty section of this syllabus. The use of these tools can also introduce unique copyright, privacy, and ethical considerations. Please review the Center for Academic Innovation's Generative AI Guidance for Learners to learn more.

Coursera Coach

During the course you may encounter Coursera Coach, a generative AI-assisted tool developed by Coursera and supported on the Coursera platform. Using the Coursera Coach is optional. Please note that you should verify AI-generated content for accuracy, The Coursera Coach should not replace your personal analysis of the content.

Jupyter Notebooks

In this course, the Jupyter Notebook tool is used to provide learners a programming environment to work in, directly on the Coursera platform through Coursera Labs. This environment utilizes Jupyter Notebooks to combine code and text.

This environment is intended to provide a ready-made workspace for learners to interact with course materials and without the need for local setup. These notebooks are available for learners to run locally, however support can only be provided for learners who utilize the Coursera Labs environment.

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